Chapter 15

Insurance Companies



Outline

- Foundations of Insurance
- Insurance Categories
- Organization of Insurance Companies
- Insurance Industry Structure
- Life Insurance Companies
- Property and Casualty Insurance Companies
- Insurance Industry Regulation
- Insurance Industry Recent Trends

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Foundations of Insurance
Major Issues



- Adverse selection occurs when the people most likely to benefit from a transaction are the ones who actively seek out the transaction and are thus likely to be selected.
  - Insured have higher risk than general population.
  - Loss probability statistics gathered for the entire population may not truly reflect the loss potential for the people who actually want to buy policies.

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Foundations of Insurance
Major Issues



- Moral hazard refers to the fact that an insurance policy changes the behaviour of the insured person.
  - A fire insurance policy written for more than the value of the property may induce the owner to arson.
  - A generous automobile insurance policy may encourage reckless driving.
- Concern arises when we cannot observe people's actions and so cannot judge whether a poor outcome is intentional or just a result of bad luck.
- Solutions to the moral hazard problem are hard to find.

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Foundations of Insurance



- To alleviate problems, insurance companies try the following:
  - information collection and screening
  - risk-based premium
  - restrictive provisions
  - prevention of fraud by investigations
  - cancellations of insurance
  - deductibles
  - co-insurance
  - limits on the insurance coverage

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Foundations of Insurance



- The insured (party covered by insurance) and the beneficiary (party receiving payout if loss occurs) must be related.
- The insurer (party selling the insurance) must be provided with complete and accurate information by the insured.
- The insured is not supposed to profit from buying insurance.
- The payout needs to be adjusted with any compensation from a third party.
- The insurer must have a pool of insured large enough to diversify the risks.
- The loss must be quantifiable.
- The insurer must be able to estimate the loss occurrence.

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Foundations of Insurance



- Risk-averse people with incentives buy insurance.
- Insurance is usually sold through agents.
  - Independent agent: sell insurance for a number of different companies (i.e., insurance broker)
  - Exclusive agent: sell insurance for only one company (i.e., insurance agent)
- Regardless of the sales approach, insurance companies hire an underwriter who decides whether to accept or reject the issuance of a policy.

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Insurance Categories



- Insurance: a device whereby an individual or a business transfers the risk of uncertain financial loss by payment of a premium.
- Broadly grouped as life insurance and property and casualty insurance with further classifications:
  - Life insurance
  - Health insurance
  - Property insurance
  - Liability insurance
  - Credit insurance
  - Farm insurance
  - Reinsurance

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Insurance Categories

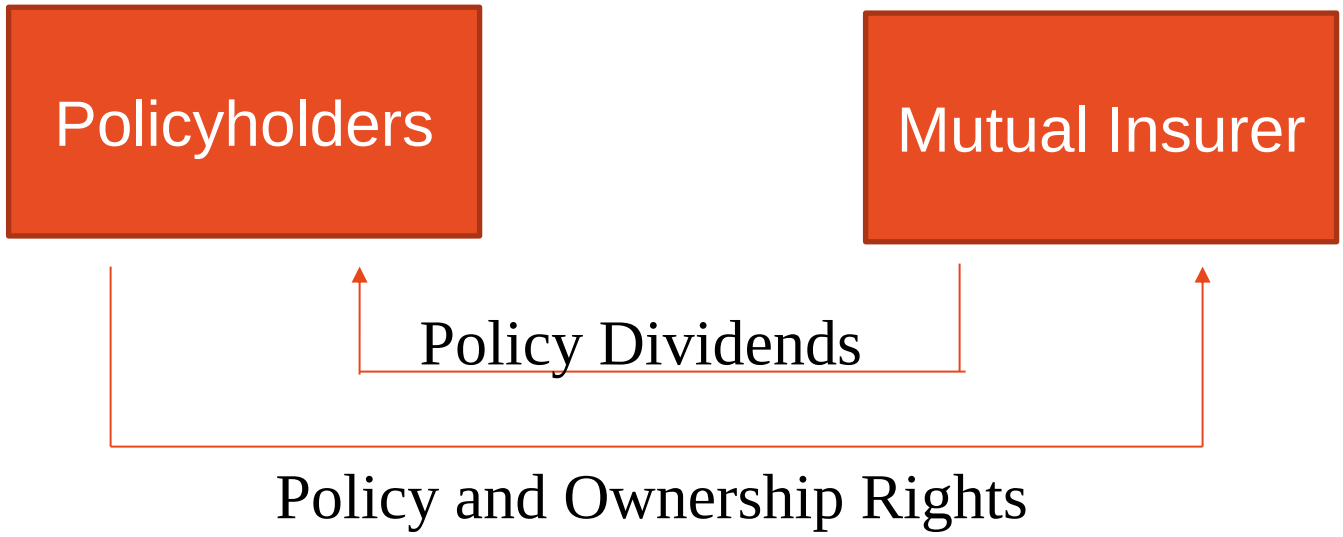
- Ten largest insurance companies (as of December 31, 2019):

Company	Total Assets
Manulife Financial Group	$809.130 billion
Great-West LifeCo	$451.167 billion
Sun Life Financial	$297.202 billion
Fairfax Financial	$92.203 billion
Industrial Alliance Financial Corp.	$73.148 billion
Desjardins Insurance	$53.152 billion
Intact Financial	$32.292 billion
E-L Financial Corp. (Empire Life)	$23.749 billion
RBC Insurance	$18.191 billion
Co-operators Group	$15.002 billion

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Organization of Insurance Companies

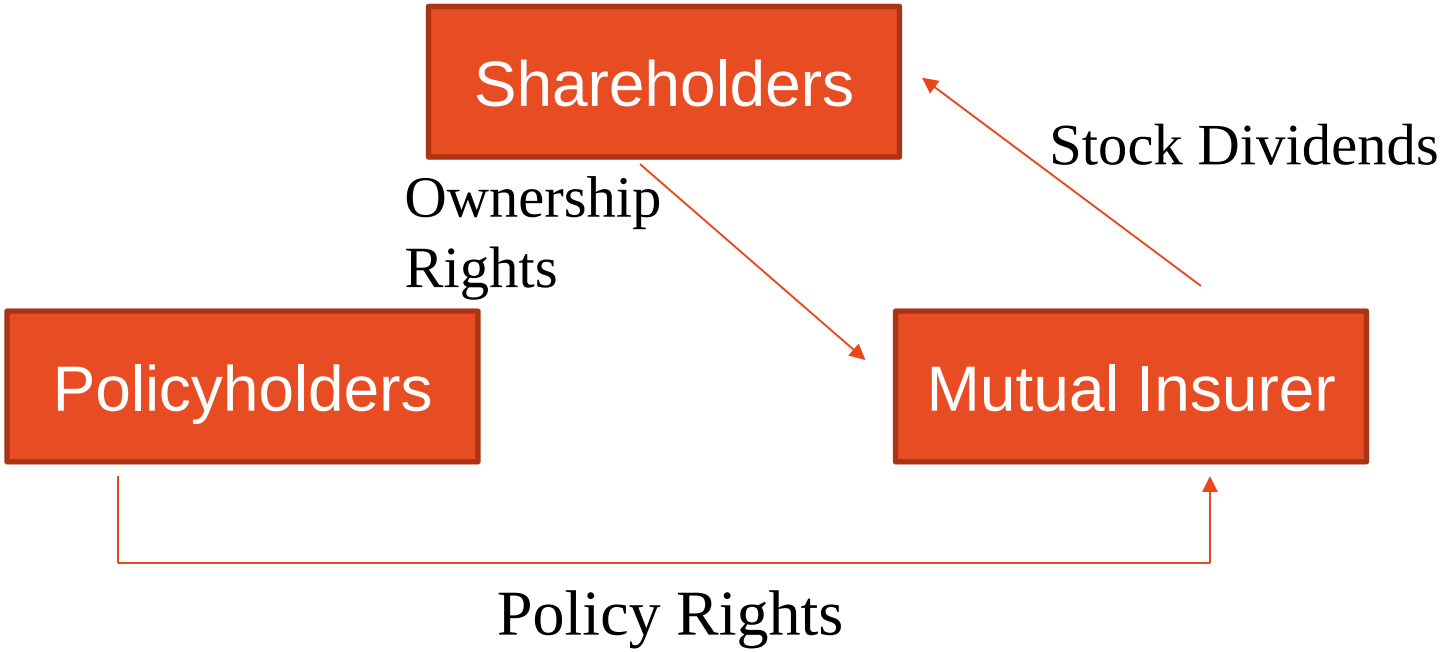
- Mutual Insurance company: accumulated profits owned by policyholders.



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Organization of Insurance Companies

- Stock Insurance company: accumulated profits owned by shareholders.



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Insurance Industry Structure



- Insurance pools risk exposures, allowing losses to be shared and reducing the uncertainty of loss.
- Insurance industry is a clunky legacy system.
- Clients have evolving expectations for insurance transactions, insurance products and services.
  - Online distribution platform for insurers with a direct-to-consumer channel.
    - Montreal-based InsurTech: Breath Life

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Insurance Industry Structure



- A healthy economy supports earnings growth at Canadian life insurers and premium growth for property and casualty firms, despite the ongoing drag of low interest rates on the insurers' profitability and investment performance.
- Solid asset quality remains a driver of the stable outlook.
- Insurers have the capacity to invest in technology to guard against competitive disruption.
  - This investment should enhance client interface and drive opportunities for all insurers to benefit from claims efficiencies and risk segmentation.

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Life Insurance Companies
Overview



- Protect nearly 29 million Canadians with various life insurance products and pay $98 billion in benefits.
  - The average life insurance protection per household is $423,000 in 2019, up from $417,000.
- The industry is competitive, with 100 Canadian insurers and 38 foreign-owned insurers.
  - 78 private life insurers, 8 not-for-profit health benefit providers and 14 fraternal benefit societies
  - Many private life insurers are diversified financial institutions offering more than life insurance.

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Life Insurance Companies
Overview



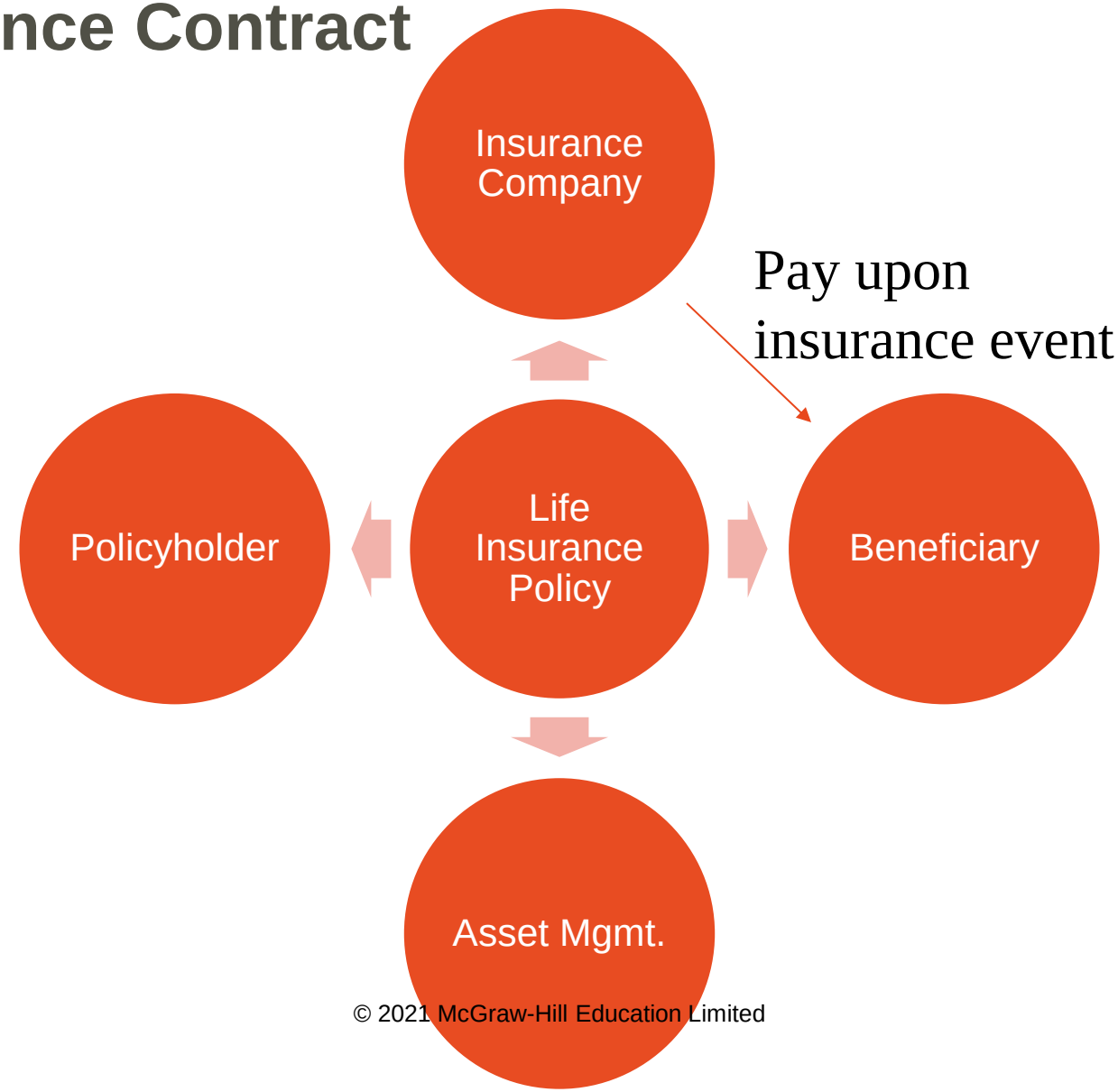
- Top six life insurers based on annual revenues (2019):

Company	Founded	A.M. Rating	Annual Revenues	Net Income
Manulife Financial	1887	A+	$77.83 B	$5,276 M
Great-West LifeCo (Canada Life)	1891	A+ (Superior)	$44.71 B	$2,492 M
Sun Life Financial	1846	A+	$39.69 B	$2,713 M
IA Financial	1892	A	$15.27 B	$708 M
RBC Insurance	1864	A	$4.27 B	$775 M
Empire Life (E-L Financial)	1923	A (Excellent)	$2.12 B	$187.44 M

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Life Insurance Companies

Insurance Contract



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Life Insurance Companies
Lines of Life Insurance: Ordinary Life



- Ordinary life insurance is sold to individuals—policyholders make periodic premium payments in exchange for coverage.
  - Term life policy is pure life insurance with expiry date and no savings element attached.
  - Permanent life policy offers a package of pure life insurance protection and investing opportunities with no expiry.
    - Whole life
    - Universal life

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Life Insurance Companies
Lines of Life Insurance: Ordinary Life



- Term life insurance
  - temporary insurance protection
  - low cost
  - no cash value
  - usually renewable
  - likely convertible to permanent life insurance without further underwriting

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Life Insurance Companies
Lines of Life Insurance: Ordinary Life



- Permanent Life Insurance
  - meet life-long protection needs
  - have an investment component
  - no expiry date as long as premiums are paid
  - more expensive to own
  - build cash value
  - loans are permitted against the policy
  - favourable tax treatment of policy earnings
  - flexible premium allocation

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Life Insurance Companies
Lines of Life Insurance: Ordinary Life



- Permanent Life insurance can be divided into two categories:
  - Whole Life: traditional form where insurer takes on all risks related to death and the underlying investment performance.
    - Premiums partly fund an increase in cash value.
    - Expensive.
  - Universal Life: policyholders select investment options and insurer assumes the risk related to death.
    - Policyholder may vary the timing and amount of premium as circumstances change.
    - Cash value depends on the pace of the premium.

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Life Insurance Companies
Lines of Life Insurance: Group and Credit



- Group life insurance covers a large number of people under a single policy.
  - Contributory: both the employer and the employee cover a share of the premiums.
  - Noncontributory: the costs are borne entirely by the employer.
- Credit life insurance protects lenders in the event a borrower dies prior to the repayment of a debt contract, such as a mortgage or car loan.

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Life Insurance Companies
Lines of Life Insurance: Other Activities



- Other activities of life insurers include the sale of annuities, private pension plans, and accident and health insurance.
  - Annuities protect against the risk of outliving your retirement money.
    - Contract is initiated by investing a lump sum or making periodic payments before the annuity payments begin.
  - Canadians' retirement savings in RRSP, TSFA and RRIF are managed by the industry.
  - Health insurance supplements government benefits.

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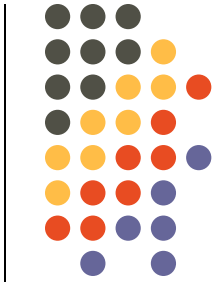
Life Insurance Companies
Assets and Liabilities



- With a steady flow of premiums and predictable benefit payments, life insurers can securely invest in long-term capital markets.
- The mix of investments within general and segregated funds reflects the nature of the underlying insurance contracts of two funds.
- Liabilities come from the premiums paid by insured clients and from pension funds under management.

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Life Insurance Companies
Assets and Liabilities



- Segregated funds are kept separate from the rest of the insurer's activities.

Assets ($762 B)	Liabilities ($665 B) and Equities ($97 B)
General Funds ($470 B)	
5% short-term investment	72% policyholder liabilities
87% long-term investment	7% other liabilities
8% other investment	21% equities and long-term debt
Segregated Funds ($292 B)	
16% cash & other investment	100% policyholder liabilities
34% real estate & mortgages	
50% mutual funds	

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Property-Casualty Insurance Companies



- Currently, 198 companies sell property-casualty (P&C) insurance in Canada, and approximately half of them are foreign firms.
  - Top 10 firms hold 65% market share.
  - In 2018, Intact Financial was the top firm, writing 14.4% of all P&C insurance premiums.
  - Most banks have insurance subsidiaries, as they are not allowed to sell insurance through their branch networks.

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Property-Casualty Insurance Companies



- Property insurance protects the insured against losses of real and personal property from fire, theft, storm, explosion and even neglect.
- The policy can be either a named-peril policy or an open-peril policy.
  - Named-peril policies cover losses associated with specific named events (e.g., flood insurance).
  - Open-peril policies insure against all perils except those specifically excluded by the policy (e.g., war)

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Property-Casualty Insurance Companies



- Casualty insurance offers protection from legal responsibility for losses stemming from damage to another's property or an injury to another person.
- Distinctions between property and casualty insurance are increasingly become blurred because they are often sold together.
  - Auto insurance: you wrecking your car is covered under property insurance, while you accidentally hitting another car with yours is covered under casualty insurance.

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Property-Casualty Insurance Companies



- Top ten leading private P&C Insurers (2019):

Company	Market share
Intact Group	14.44%
Aviva Group	8.91%
Desjardins Group	8.33%
Lloyd's Underwriters	5.91%
Cooperators Group	5.57%
Wawanesa Mutual Insurance Company	5.47%
TD Insurance Group	5.39%
RSA Group	5.13%
Economical Group	4.10%
Northbridge Group	2.88%

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P&C Insurance Companies
Lines of P&C Insurance



- Insurance companies earn revenues through premiums and on investments they make.
- Report premiums in two ways:
  - Direct written premiums: total premiums collected
  - Net written premiums: total amounts adjusted for portions paid to reinsurers
- To spread the risk, a reinsurance company agrees to accept risks of another insurance company in exchange for a payment.

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P&C Insurance Companies
Lines of P&C Insurance



- P&C insurers have $59.6 billion in direct written premiums on six major lines in 2019.
- Among the net written premiums of $54.1 billion:
  - Automobile Insurance: 44.1%
  - Property Insurance:
    - Personal: 22.2%
    - Commercial:14.2%
  - Liability Insurance: 10.2%
  - Specialized Insurance: 7.1%
  - Accident and Sickness Insurance: 2.2%

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P&C Insurance Companies
Lines of P&C Insurance

- Total direct claim costs by line in 2019:

Line of Business	$ million	% of the total claims
Auto	18,205	44.6
Personal property	7,810	20.0
Commercial property	6,466	16.5
Liabilities	3,746	9.6
Specialized	1,985	5.1
Accident and sickness	871	2.2
	$39,084	100%

- Net claims incurred are the total claim costs less any share to be paid by reinsurers.

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P&C Insurance Companies
Assets and Liabilities



- Assets
  - Out of the total investment of $117 billion, many of those assets have fixed returns that can be liquidated easily.

Assets	Amount	Fraction
Bonds	$80,586 M	68.9%
Stocks	$10,904 M	9.3%
Mortgages	$1,202 M	1.0%
Real estates	$530 M	0.5%
Term deposits	$4,436 M	3.8%
Others	$19,366 M	16.5%

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P&C Insurance Companies
Assets and Liabilities



- Liabilities
  - About 57% of the total liabilities and equity are unpaid claims and adjustment expenses items.
  - Risk exposure of property insurance is limited to the property value, but liability risk exposure is much more difficult to determine.
  - Liability risk exposure can have long lag times (tails).
  - Liability lines may be subject to social inflation.

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P&C Insurance Companies
Assets and Liabilities



- To determine insurance premiums, insurers consider the likelihood of a customer making a claim and how much those claims will like cost.
- Underwriting risk is the risk that premiums are insufficient to cover losses and operating expenses after taking into account investment income.
- Three sources of underwriting risk:
  - unexpected increases in loss rates
  - unexpected increases in expenses
  - unexpected decreases in investment yields

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P&C Insurance Companies
Assets and Liabilities



- Four key ratios:
  - Loss ratio: the actual claims incurred on a business line relative to the net premium earned
  - Expense ratio: the expenses incurred relative to the net premium written, including loss adjustment expenses and mainly employee compensations
  - Combined ratio: includes both the loss ratio and the expense ratio
  - Operating ratio: measures overall profitability by subtracting the investment yield from the combined ratio
    - Investment yield is measured as net interest income divided by premiums earned

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P&C Insurance Companies
Assets and Liabilities



- P&C industry ratios in 2019:
  - earned loss ratio = 67.7%
  - expense ratio = 30.9%
  - combined ratio = 98.6%
  - investment yield = 2% lowest on record
  - operating ratio = 98.6% − 2% = 96.6%
- Overall profitability = 100% − 96.6% = 3.4%
  - Less profitable firms exit the industry, and there is a rapid increase in premiums among the survivors.

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Insurance Industry Regulation
Life Insurance Companies



- Federal and provincial governments share jurisdiction.
- OSFI: governs through the Insurance Companies Act
  - prudential regulation, concerned with safety and soundness of the industry
- Provincial regulation
  - marketing of insurance products and licensing requirements
- Canadian Life and Health Insurance Compensation Corporation: industry-funded organization to protect policyholders
  - $200,000 on claims in case of bankruptcy
  - $60,000 in cash policy coverage.

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Insurance Industry Regulation
P&C Insurance Companies



- As P&C insurers may be federally or provincially incorporated, regulation is shared by provincial and federal government.
- OSFI is the federal regulator responsible for 75% of companies.
- Property and Casualty Insurance Compensation Corporation: pay outstanding claims and refund premiums paid in advance up to the following limits:
  - auto and commercial insurance policies: up to $250,000
  - home insurance policies: up to $300,000 per policy

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Insurance Industry Recent Trends
Major Issues



- Insurance industry becoming more global
  - cross-country mergers (insurance companies as well as universal banks)
- Insurance crime
- Severe weather claims
- Catastrophic losses
- Introduction and acceleration of insurance market reforms
  - auto insurance

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Insurance Industry Recent Trends
InsurTech



- Insurers are increasingly focused on consumers' needs with more tailored product offerings.
- Despite lower premium growth and rising claims costs, insurance companies continue to innovate and adapt through the following:
  - artificial Intelligence (AI)
  - robotic process automation (RPA)
  - machine learning

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Insurance Industry Recent Trends
InsurTech



- Insurance companies must resolve the “synthesis challenge”—integrating innovation into the legacy resistant environment.
- Many are beginning to pivot from investments that support business as usual to financing innovations that facilitate more fundamental business model changes.
- All need to adapt a more customer-centric approach to run their businesses.

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